

Vector Spaces

Definition

A vector space V is a set of vectors with two operations defined, addition and scalar multiplication, which satisfy the following axioms for all $\vec{u}$, $\vec{v}$, and $\vec{w} \in V$ and for all $a, b \in \mathbb{R}$.

$$\text{I} \bullet \vec{u} + \vec{v} \in V.$$

$$\text{II} \bullet \vec{u} + \vec{v} = \vec{v} + \vec{u}$$

$$\text{III} \bullet \text{There exists } \vec{0} \in V \text{ such that } \vec{0} + \vec{u} = \vec{u}.$$

$$\text{IV} \bullet \text{There exists } -\vec{u} \in V \text{ such that } \vec{u} + (-\vec{u}) = \vec{0}. \quad \text{V} \bullet (\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

$$\text{VI} \bullet a\vec{u} \in V.$$

$$\text{VII} \bullet a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}.$$

$$\text{VIII} \bullet (a + b)\vec{u} = a\vec{u} + b\vec{u}.$$

$$\text{IX} \bullet a(b\vec{u}) = (ab)\vec{u}.$$

$$\text{X} \bullet 1\vec{u} = \vec{u}.$$

Example — Vector Space of Polynomials

Let $\mathbb{P}_2$ be the set of all polynomials of at most degree 2 as well as the zero polynomial. Define addition to be the standard addition of polynomials, and scalar multiplication the usual multiplication of a polynomial by a number. Then $\mathbb{P}_2$ is a vector space.

Solution. We can write $\mathbb{P}_2$ explicitly as

$$\mathbb{P}_2 = \{a_2x^2 + a_1x + a_0 \mid a_i \in \mathbb{R} \text{ for all } i\}$$

To show that $\mathbb{P}_2$ is a vector space, we verify the axioms. Let $p(x), q(x), r(x)$ be polynomials in $\mathbb{P}_2$ and let a, b, c be real numbers. Write $p(x) = p_2x^2 + p_1x + p_0$, $q(x) = q_2x^2 + q_1x + q_0$, and $r(x) = r_2x^2 + r_1x + r_0$.

- We first prove that addition of polynomials in $\mathbb{P}_2$ is closed. For two polynomials in $\mathbb{P}_2$ we need to show that their sum is also a polynomial in $\mathbb{P}_2$. From the definition of $\mathbb{P}_2$, a polynomial is contained in $\mathbb{P}_2$ if it is of degree at most 2 or the zero polynomial.

$$\begin{aligned} p(x) + q(x) &= p_2x^2 + p_1x + p_0 + q_2x^2 + q_1x + q_0 \\ &= (p_2 + q_2)x^2 + (p_1 + q_1)x + (p_0 + q_0) \end{aligned}$$

The sum is a polynomial of degree 2 and therefore is in $\mathbb{P}_2$. It follows that $\mathbb{P}_2$ is closed under addition.

- We need to show that addition is commutative, that is $p(x) + q(x) = q(x) + p(x)$.

$$\begin{aligned} p(x) + q(x) &= p_2x^2 + p_1x + p_0 + q_2x^2 + q_1x + q_0 \\ &= (p_2 + q_2)x^2 + (p_1 + q_1)x + (p_0 + q_0) \\ &= (q_2 + p_2)x^2 + (q_1 + p_1)x + (q_0 + p_0) \\ &= q_2x^2 + q_1x + q_0 + p_2x^2 + p_1x + p_0 \\ &= q(x) + p(x) \end{aligned}$$

- Next, we need to show that addition is associative. That is, that $(p(x) + q(x)) + r(x) = p(x) + (q(x) + r(x))$.

$$\begin{aligned} (p(x) + q(x)) + r(x) &= (p_2x^2 + p_1x + p_0 + q_2x^2 + q_1x + q_0) + r_2x^2 + r_1x + r_0 \\ &= (p_2 + q_2)x^2 + (p_1 + q_1)x + (p_0 + q_0) + r_2x^2 + r_1x + r_0 \\ &= (p_2 + q_2 + r_2)x^2 + (p_1 + q_1 + r_1)x + (p_0 + q_0 + r_0) \\ &= p_2x^2 + p_1x + p_0 + (q_2 + r_2)x^2 + (q_1 + r_1)x + (q_0 + r_0) \\ &= p_2x^2 + p_1x + p_0 + (q_2x^2 + q_1x + q_0 + r_2x^2 + r_1x + r_0) \\ &= p(x) + (q(x) + r(x)) \end{aligned}$$

- Next, we must prove that there exists an additive identity. Let $0(x) = 0x^2 + 0x + 0$.

$$p(x) + 0(x) = p_2x^2 + p_1x + p_0 + 0x^2 + 0x + 0$$

$$\begin{aligned}
&= (p_2 + 0)x^2 + (p_1 + 0)x + (p_0 + 0) \\
&= p_2x^2 + p_1x + p_0 \\
&= p(x)
\end{aligned}$$

Hence an additive identity exists, specifically the zero polynomial.

- Next we must prove that there exists an additive inverse. Let $-p(x) = -p_2x^2 - p_1x - p_0$ and consider the following:

$$\begin{aligned}
p(x) + (-p(x)) &= p_2x^2 + p_1x + p_0 + (-p_2x^2 - p_1x - p_0) \\
&= (p_2 - p_2)x^2 + (p_1 - p_1)x + (p_0 - p_0) \\
&= 0x^2 + 0x + 0 \\
&= 0(x)
\end{aligned}$$

Hence an additive inverse $-p(x)$ exists such that $p(x) + (-p(x)) = 0(x)$.

We now need to verify the axioms related to scalar multiplication.

- First we prove that $\mathbb{P}_2$ is closed under scalar multiplication. That is, we show that $ap(x)$ is also a polynomial of degree at most 2.

$$ap(x) = a(p_2x^2 + p_1x + p_0) = ap_2x^2 + ap_1x + ap_0$$

Therefore $\mathbb{P}_2$ is closed under scalar multiplication.

- We need to show that $a(p(x) + q(x)) = ap(x) + aq(x)$.

$$\begin{aligned}
a(p(x) + q(x)) &= a(p_2x^2 + p_1x + p_0 + q_2x^2 + q_1x + q_0) \\
&= a((p_2 + q_2)x^2 + (p_1 + q_1)x + (p_0 + q_0)) \\
&= a(p_2 + q_2)x^2 + a(p_1 + q_1)x + a(p_0 + q_0) \\
&= (ap_2 + aq_2)x^2 + (ap_1 + aq_1)x + (ap_0 + aq_0) \\
&= ap_2x^2 + ap_1x + ap_0 + aq_2x^2 + aq_1x + aq_0 \\
&= ap(x) + aq(x)
\end{aligned}$$

- Next we show that $(a + b)p(x) = ap(x) + bp(x)$.

$$\begin{aligned}
(a + b)p(x) &= (a + b)(p_2x^2 + p_1x + p_0) \\
&= (a + b)p_2x^2 + (a + b)p_1x + (a + b)p_0 \\
&= ap_2x^2 + ap_1x + ap_0 + bp_2x^2 + bp_1x + bp_0 \\
&= ap(x) + bp(x)
\end{aligned}$$

- The next axiom which needs to be verified is $a(bp(x)) = (ab)p(x)$.

$$\begin{aligned}
a(bp(x)) &= a(b(p_2x^2 + p_1x + p_0)) \\
&= a(bp_2x^2 + bp_1x + bp_0) \\
&= abp_2x^2 + abp_1x + abp_0 \\
&= (ab)(p_2x^2 + p_1x + p_0) \\
&= (ab)p(x)
\end{aligned}$$

- Finally, we show that $1p(x) = p(x)$.

$$\begin{aligned}
1p(x) &= 1(p_2x^2 + p_1x + p_0) \\
&= 1p_2x^2 + 1p_1x + 1p_0 \\
&= p_2x^2 + p_1x + p_0 \\
&= p(x)
\end{aligned}$$

Since the above axioms hold, we know that $\mathbb{P}_2$ as described above is a vector space.

